

លីមីតនៃអនុគមន៍

១. និយមន័យ

បើ x ខិតជិត a ហើយអនុគមន៍ $y = f(x)$ ខិតជិតតម្លៃ b ណាមួយនោះគេកំណត់សរសេរ

$$\lim_{x \rightarrow a} f(x) = b \text{ ។}$$

២. ប្រមាណលីមីតលីមីត

បើ $\lim_{x \rightarrow a} f(x) = L, \lim_{x \rightarrow a} g(x) = M, \lim_{x \rightarrow a} h(x) = N$ ហើយ L, M, N ជាចំនួនពិតយើងបាន៖

$$\text{ក. } \lim_{x \rightarrow a} k = k$$

$$\text{ខ. } \lim_{x \rightarrow a} [f(x) \pm g(x)] = L \pm M$$

$$\text{គ. } \lim_{x \rightarrow a} kf(x) = k \lim_{x \rightarrow a} f(x) = kL$$

$$\text{ឃ. } \lim_{x \rightarrow a} [f(x) + g(x) - h(x)] = L + M - N$$

$$\text{ង. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}, M \neq 0, g(x) \neq 0$$

$$\text{ច. } \lim_{x \rightarrow a} [f(x) \times g(x) \times h(x)] = L \times M \times N$$

ការគណនាលីមីត

ដើម្បីគណនា $\lim_{x \rightarrow x_0} f(x)$ គេជំនួស $x = x_0$ ទៅក្នុងកន្សោមនៃអនុគមន៍ $f(x)$ នោះមានន័យ

ថារក $f(x_0)$ ។

- បើ $f(x_0) = k, k \in \mathbb{R}$ នោះគេថា k ជាលីមីតនៃអនុគមន៍ $f(x)$ ។
- បើ $f(x_0)$ ស្ថិតនៅក្នុងទម្រង់ $\frac{0}{0}, \frac{\infty}{\infty}, +\infty - \infty, 0 \times \infty, 1^\infty, \infty^0, 0^0$ នោះកន្សោមទាំងនេះ មានរាងមិនកំណត់ ហើយគេប្រើវិធីផ្សេងទៀត ដើម្បីគណនាលីមីតនេះ ។

អនុវត្ត ១៖

1. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\text{ក. } \lim_{x \rightarrow 1} (x^3 + x^2 - 5)$$

$$\text{ខ. } \lim_{x \rightarrow 0} \frac{x^3 - 3x + 4}{x + 1}$$

$$\text{គ. } \lim_{x \rightarrow \frac{1}{2}} \frac{x - 1}{x + 1}$$

$$\text{ឃ. } \lim_{x \rightarrow \frac{3}{2}} (x - 1)(x^2 + 1)$$

ង. $\lim_{x \rightarrow \pi} [\cos 2x - \sin 2x]$

ច. $\lim_{x \rightarrow 0} \frac{2x - 1}{x^2 + 2x + 5}$

ឆ. $\lim_{x \rightarrow 3} (3 \ln x - \ln 3)$

ជ. $\lim_{x \rightarrow 2} \frac{e^x - 1}{2x - 1}$

ឈ. $\lim_{x \rightarrow 2} x^2 \sqrt{x^2 + 5x + 2}$

ញ. $\lim_{x \rightarrow 2} x \sqrt{x + 4} \sqrt[3]{x - 6}$

2. គេឱ្យ $\lim_{x \rightarrow a} f(x) = 4$, $\lim_{x \rightarrow a} g(x) = 2$ ។ គណនាតម្លៃនៃលីមីតខាងក្រោម៖

ក. $\lim_{x \rightarrow a} [5f(x) + 6g(x)]$

ខ. $\lim_{x \rightarrow a} [f(x)]^3$

គ. $\lim_{x \rightarrow a} \frac{1}{g(x)}$

ឃ. $\lim_{x \rightarrow a} \sqrt{\frac{f(x)}{g(x)}}$

ង. $\lim_{x \rightarrow a} \frac{f(x)}{f(x) - 2g(x)}$

ច. $\lim_{x \rightarrow a} xf(x)g(x)$

ឆ. $\lim_{x \rightarrow a} \frac{[f(x)]^2 - 4[g(x)]^2}{f(x) - 2g(x)}$

ជ. $\lim_{x \rightarrow a} \frac{6x + 3}{xf(x) + g(x)} ; a \neq -\frac{1}{2}$

៣. លីមីតរាងមិនកំណត់

- លីមីតរាងមិនកំណត់ $\frac{0}{0}$

ដើម្បីគណនាលីមីត $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ រាងមិនកំណត់ $\frac{0}{0}$ គេត្រូវអនុវត្តតាមវិធានដូចខាងក្រោម៖

- បំបែកភាគយក និងភាគបែងជាផលគុណកត្តា ដែលមាន $x - a$ ជាកត្តារួម
- សម្រួលកត្តារួម $(x - a)$ ចោល
- រួចគណនាលីមីតនៃប្រភាគថ្មី (ដោយជំនួស x ដោយ a) ។

សង្ខេប

- បើ $x \rightarrow a$ នោះកត្តាត្រូវសម្រួលចោល $(x - a)$ ។
- បើ $x \rightarrow -a$ នោះកត្តាត្រូវសម្រួលចោល $(x + a)$ ។
- បើ $x \rightarrow 0$ នោះកត្តាត្រូវសម្រួលចោល x ។

រូបមន្តសំខាន់ៗមួយចំនួន

- $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

ឧទាហរណ៍: $a^2 - b^2 = (a - b)(a + b)$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^4 - b^4 = (a - b)(a^3 + a^2b + ab^2 + b^3)$$

$$a^{100} - b^{100} = (a - b)(a^{99} + a^{98}b + \dots + ab^{99})$$

- $a^n - 1 = (a - 1)(a^{n-1} + a^{n-2} + \dots + a + 1)$

ឧទាហរណ៍: $a^2 - 1 = (a - 1)(a + 1)$

$$a^3 - 1 = (a - 1)(a^2 + a + 1)$$

$$a^{100} - 1 = (a - 1)(a^{99} + a^{98} + \dots + 1)$$

- $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots + b^{n-1})$

ដែល n ជាចំនួនគត់សេស

- $a^n + 1 = (a + 1)(a^{n-1} - a^{n-2} + a^{n-3} - \dots + 1)$

ដែល n ជាចំនួនគត់សេស

កន្សោមឆ្លាស់

- $(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}) = \sqrt{a}^2 - \sqrt{b}^2 = a - b$

- $(\sqrt{a} - b)(\sqrt{a} + b) = \sqrt{a}^2 - b^2 = a - b^2$

- $(a - \sqrt{b})(a + \sqrt{b}) = a^2 - \sqrt{b}^2 = a^2 - b$

កន្សោមបំពេញ

- $(\sqrt[3]{a} - \sqrt[3]{b})(\sqrt[3]{a^2} + \sqrt[3]{ab} + \sqrt[3]{b^2}) = \sqrt[3]{a}^3 - \sqrt[3]{b}^3 = a - b$

- $(\sqrt[3]{a} + \sqrt[3]{b})(\sqrt[3]{a^2} - \sqrt[3]{ab} + \sqrt[3]{b^2}) = \sqrt[3]{a}^3 + \sqrt[3]{b}^3 = a + b$

លំហាត់គម្រោង

គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 5} \frac{x^2 - 25}{2x - 10}$ រាងមិនកំណត់ $\frac{0}{0}$

គេបាន៖ $\lim_{x \rightarrow 5} \frac{x^2 - 25}{2x - 10} = \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{(x-5)2} = \lim_{x \rightarrow 5} \frac{x+5}{2} = \frac{5+5}{2} = \boxed{5}$

ខ. $\lim_{x \rightarrow -7} \frac{3x + 21}{x^2 - 49}$ រាងមិនកំណត់ $\frac{0}{0}$

គេបាន៖ $\lim_{x \rightarrow -7} \frac{3x + 21}{x^2 - 49} = \lim_{x \rightarrow -7} \frac{(x+7)3}{(x+7)(x-7)} = \lim_{x \rightarrow -7} \frac{3}{x-7} = \frac{3}{-7-7} = \boxed{-\frac{3}{14}}$

គ. $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - x - 6}$ រាងមិនកំណត់ $\frac{0}{0}$

គេបាន៖ $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - x - 6} = \lim_{x \rightarrow 3} \frac{(x-3)(x-1)}{(x-3)(x+2)} = \lim_{x \rightarrow 3} \frac{x-1}{x+2} = \frac{3-1}{3+2} = \boxed{\frac{2}{5}}$

ប្រសិនបើលីមីតមានរាងមិនកំណត់ $\frac{0}{0}$

ហើយភាគយក និងភាគបែងជាពហុធា
ដ៏ក្រេឡិច(រៀបតាមលំដាប់) យើងគ្រាន់
តែ គុណតម្លៃលេខទៅនឹងកត្តាដែលចាប់
ហើយសាមញ្ញភាពតួដើម និងតួចុង
ផ្ទៀងផ្ទាត់ ។

$$x^2 = x \times x$$

$$-6 = (-3) \times (2)$$

ឃ. $\lim_{x \rightarrow 3} \frac{2x^3 - 5x^2 - 8x + 15}{x^2 - 9}$ រាងមិនកំណត់ $\frac{0}{0}$

$$\begin{aligned}\text{គេបាន: } \lim_{x \rightarrow 3} \frac{2x^3 - 5x^2 - 8x + 15}{x^2 - 9} &= \lim_{x \rightarrow 3} \frac{(x-3)(2x^2 + x - 5)}{(x-3)(x+3)} \\ &= \lim_{x \rightarrow 3} \frac{2x^2 + x - 5}{x+3} \\ &= \frac{2 \times 3^2 + 3 - 5}{3+3} \\ &= \frac{18 + 3 - 5}{3+3} \\ &= \frac{16}{6} = \frac{8}{3}\end{aligned}$$

ពេល $x \rightarrow 3$ ត្រូវចាប់ $x-3$

ជាកត្តា ដើម្បីសម្រួល

ភាគយក $2x^3 - 5x^2 - 8x + 15$

ហេតុនេះយើងត្រូវយកពហុធានេះ

ចែកនឹង $(x-3)$

ដើម្បីដឹងថា $(x-3)$ គុណនឹង

កន្សោមអ្វី

Long Division

$$\begin{array}{r} 2x^2 + x - 5 \\ x-3 \overline{) 2x^3 - 5x^2 - 8x + 15} \\ \underline{2x^3 - 6x^2} \\ x^2 - 8x \\ \underline{x^2 - 3x} \\ -5x + 15 \\ \underline{-5x + 15} \\ \text{Remainder } 0 \end{array}$$

Synthetic Division

$$\begin{array}{r|rrrrr} 3 & 2 & -5 & -8 & 15 & \\ & & 6 & 3 & 15 & \\ \hline & 2 & 1 & -5 & 0 & \end{array}$$

Remainder

អនុវត្ត ២៖

3. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

ខ. $\lim_{x \rightarrow 9} \frac{x^2 - 81}{x - 9}$

គ. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{4 - x}$

ឃ. $\lim_{x \rightarrow -6} \frac{x^2 - 36}{x + 6}$

ង. $\lim_{x \rightarrow -2} \frac{x^2 - 4}{2 + x}$

ច. $\lim_{x \rightarrow -11} \frac{x^2 - 121}{x + 11}$

4. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 1} \frac{3x^2 - 4x + 1}{x - 1}$

ខ. $\lim_{x \rightarrow 4} \frac{3x^2 + 2x - 56}{x - 4}$

គ. $\lim_{x \rightarrow 7} \frac{x^2 + x - 56}{x - 7}$

ឃ. $\lim_{x \rightarrow 6} \frac{x^2 + x - 42}{6x - 36}$

$$ង. \lim_{x \rightarrow -1} \frac{-x^2 + 2x + 3}{x^2 - 1}$$

$$ច. \lim_{x \rightarrow 3} \frac{4x^2 + 2x - 42}{10x - 30}$$

$$ឆ. \lim_{x \rightarrow 3} \frac{2x^2 + 6x - 36}{x - 3}$$

$$ជ. \lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{2x + 4}$$

$$ឈ. \lim_{x \rightarrow 1} \frac{5x^2 + 5x - 10}{2x - 2}$$

$$ញ. \lim_{x \rightarrow 3} \frac{5x^2 + x - 48}{3x - 9}$$

$$ដ. \lim_{x \rightarrow 5} \frac{2x^2 + 2x - 60}{x^2 + 3x - 40}$$

$$ប. \lim_{x \rightarrow 4} \frac{x^2 + 4x - 32}{x^2 - 16}$$

$$ខ. \lim_{x \rightarrow 6} \frac{2x^2 + x - 78}{100x - 600}$$

$$ឈ. \lim_{x \rightarrow 1} \frac{2x^2 + 3x - 5}{x^2 + 4x - 5}$$

$$ណ. \lim_{x \rightarrow -3} \frac{x^2 - 3x - 18}{x^2 - 9}$$

$$ក. \lim_{x \rightarrow -5} \frac{x^2 - 25}{x^2 + 3x - 10}$$

5. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow 1} \frac{2x^3 - 5x^2 + 2x + 1}{x^2 - 1}$$

$$ខ. \lim_{x \rightarrow -2} \frac{2x^3 + 7x^2 + 10x + 8}{3x^3 + 6x^2 + 4x + 8}$$

$$គ. \lim_{x \rightarrow -1} \frac{2x^3 + 5x^2 + 4x + 1}{x^3 + x^2 - x - 1}$$

$$ឃ. \lim_{x \rightarrow 1} \frac{x^4 - x^3 - x + 1}{x^3 - 5x^2 + 7x - 3}$$

$$ង. \lim_{x \rightarrow -2} \frac{5x^3 + 4x^2 - 5x + 14}{x^2 + x - 2}$$

$$ច. \lim_{x \rightarrow 1} \frac{x^3 - 3x + 2}{x^4 - 4x + 3}$$

6. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x - 1}$$

$$ខ. \lim_{x \rightarrow 2} \frac{\sqrt{3x-2} - 2}{x - 2}$$

$$គ. \lim_{x \rightarrow 5} \frac{\sqrt{4x+5} - 5}{x^2 - 25}$$

$$ឃ. \lim_{x \rightarrow 7} \frac{\sqrt{5x+14} - 7}{7 - x}$$

$$ង. \lim_{x \rightarrow -1} \frac{\sqrt{8x+9} + x}{x^2 - 2x - 3}$$

$$ច. \lim_{x \rightarrow 4} \frac{\sqrt{6x+1} - 5}{x^2 + 5x - 36}$$

$$ឆ. \lim_{x \rightarrow -4} \frac{\sqrt{-x+5} - 3}{x^2 - 16}$$

$$ជ. \lim_{x \rightarrow 4} \frac{\sqrt{10x+9} - 7}{x^2 + 10x - 56}$$

$$\text{ឈ. } \lim_{x \rightarrow 9} \frac{\sqrt{10x+10} - 10}{x^2 + x - 90}$$

$$\text{ញ. } \lim_{x \rightarrow 4} \frac{4x - x^2}{2 - \sqrt{x}}$$

$$\text{ដ. } \lim_{x \rightarrow 5} \frac{\sqrt{x} - \sqrt{5}}{x^2 + 3x - 40}$$

$$\text{ប. } \lim_{x \rightarrow 4} \frac{x^2 + 4x - 32}{\sqrt{2+x} - \sqrt{6}}$$

$$\text{ខ. } \lim_{x \rightarrow 5} \frac{4x^2 + 10x - 150}{\sqrt{30-x} - 5}$$

$$\text{ឈ. } \lim_{x \rightarrow 6} \frac{\sqrt{2x+15} - 3\sqrt{3}}{6x - 36}$$

$$\text{ណ. } \lim_{x \rightarrow 6} \frac{2x^2 + x - 78}{\sqrt{x+3} - 3}$$

$$\text{ក. } \lim_{x \rightarrow -2} \frac{x+2}{\sqrt{x^2+5} - 3}$$

7. ដោយប្រើនូវតម្លៃ $\lim_{x \rightarrow 1} \frac{x^{100} - 1}{x - 1} = 100$ ចូរគណនាតម្លៃនៃលីមីតដូចជា៖

$$\text{ក. } \lim_{x \rightarrow 1} \frac{x^{100} - 1}{x^2 - 1}$$

$$\text{ខ. } \lim_{x \rightarrow 1} \frac{x^{50} - 1}{x - 1}$$

$$\text{គ. } \lim_{x \rightarrow 1} \frac{(x^{100} - 1)^2}{(x - 1)^2}$$

• លីមីតរាងមិនកំណត់ $\frac{\infty}{\infty}$

ដើម្បីគណនាលីមីតរាងមិនកំណត់ $\frac{\infty}{\infty}$ គេត្រូវអនុវត្តតាមវិធានដូចខាងក្រោម៖

- ទាញយកតួដ៏ក្រខ្ពស់ជាងគេ ទាំងភាគយក និងភាគបែង
- សម្រួលកត្តារួមចោល រួចគណនាលីមីតនៃប្រភាគថ្មី ។

ចំណាំ៖ $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

អនុវត្ត ៣៖

8. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\text{ក. } \lim_{x \rightarrow +\infty} \frac{x+2}{x-4}$$

$$\text{ខ. } \lim_{x \rightarrow +\infty} \frac{18x-9}{100x+19}$$

$$\text{គ. } \lim_{x \rightarrow +\infty} \frac{50x+90}{20x+9}$$

$$\text{ឃ. } \lim_{x \rightarrow +\infty} \frac{x^2+x-42}{2x^2-6x-36}$$

$$ង. \lim_{x \rightarrow +\infty} \frac{9x + 90}{x - 3}$$

$$ន. \lim_{x \rightarrow +\infty} \frac{5x^2 + 5x - 10}{7x^2 + 2x - 2}$$

$$ឈ. \lim_{x \rightarrow -\infty} \frac{x^2 - 2x + 3}{x^2 - 1}$$

$$ដ. \lim_{x \rightarrow +\infty} \frac{2x^2 + 2x - 60}{x^3 - x^2 + 3x - 40}$$

$$ខ. \lim_{x \rightarrow -\infty} \frac{5x^2 + x - 78}{100x^2 - 600}$$

$$ណ. \lim_{x \rightarrow +\infty} \frac{4x^3 - 3x - 18}{x^2 - 9}$$

$$ច. \lim_{x \rightarrow +\infty} \frac{6x + 8}{200x + 11}$$

$$ជ. \lim_{x \rightarrow -\infty} \frac{5x^2 + x - 48}{x^2 + 3x - 9}$$

$$ញ. \lim_{x \rightarrow \infty} \frac{4x^3 + 2x - 42}{10x - 30}$$

$$ប. . \lim_{x \rightarrow +\infty} \frac{x^2 + 4x - 32}{x - 16}$$

$$ឆ. \lim_{x \rightarrow +\infty} \frac{2x^2 + 3x - 5}{-x^2 + 4x - 5}$$

$$ត. \lim_{x \rightarrow -\infty} \frac{4x^2 - 2x + 3}{x - 5}$$

9. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម:

$$ក. \lim_{x \rightarrow +\infty} \frac{x^2 + 10x + 5}{9x^2 + 10x + 8}$$

$$គ. \lim_{x \rightarrow +\infty} \frac{x^5 + x^4 + 7x^3 + x + 7}{19x^5 + x^3 + 8}$$

$$ង. \lim_{x \rightarrow +\infty} \frac{10x^2 + 19x + 1}{x^3 + 10x^2 + 5}$$

$$ន. \lim_{x \rightarrow +\infty} \frac{(1 - 2x)(x + 3)(4x + 1)}{x^3 - 8}$$

$$ឈ. \lim_{x \rightarrow +\infty} \left(\frac{x^3}{3x^2 - 4} - \frac{x^2}{3x + 2} \right)$$

$$ខ. \lim_{x \rightarrow +\infty} \frac{3x^2 + 100x + 11}{20x^2 + 80x + 90}$$

$$ឃ. \lim_{x \rightarrow +\infty} \frac{x^3 + 10x + 3}{x^3 + 10x^2 + 8}$$

$$ច. \lim_{x \rightarrow +\infty} \frac{7x^2 - 3x + 1}{9x^2 - 1}$$

$$ជ. \lim_{x \rightarrow +\infty} \frac{(4x + 3)^3 (2x + 1)^4}{1 + 32x^7}$$

$$ញ. \lim_{x \rightarrow -\infty} \frac{(2x^2 + 1)^3}{x^4 + x - 1}$$

10. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម:

$$ក. \lim_{x \rightarrow +\infty} \frac{x^2 + 2x + e^x}{x^2 + e^x + 1}$$

$$គ. \lim_{x \rightarrow +\infty} \frac{3x^2 + e^x + 1}{x^2 + e^x + 1}$$

$$ខ. \lim_{x \rightarrow +\infty} \frac{10x^2 + 2e^x + x}{7x^2 + e^2 + 1}$$

$$ឃ. \lim_{x \rightarrow +\infty} \frac{3e^{2x} - e^x + x^2}{5x^2 - e^x + 1}$$

$$ង. \lim_{x \rightarrow +\infty} \frac{10x^2 + 5e^x + 1}{x^3 + 10e^{2x} + 5}$$

$$ឆ. \lim_{x \rightarrow +\infty} \frac{x^2 - \ln x}{x - \ln x}$$

$$ឈ. \lim_{x \rightarrow +\infty} \frac{9x^3 - x + \ln x}{x^2}$$

$$ច. \lim_{x \rightarrow +\infty} \frac{7x^2 - 3x + 1}{9x^2 - \ln x}$$

$$ជ. \lim_{x \rightarrow +\infty} \frac{e^x + 2 \ln x}{x^2 - 4}$$

$$ញ. \lim_{x \rightarrow -\infty} \frac{4x - x^5}{2x^4 + x^5}$$

11. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow +\infty} \frac{\sqrt{2x^4 + x^2 - 1}}{1 - 3x}$$

$$គ. \lim_{x \rightarrow -\infty} \frac{2|x| + 3}{\sqrt{x^2 + x + x + 5}}$$

$$ង. \lim_{x \rightarrow +\infty} \frac{\sqrt{3x^2 - 3x + 4}}{\sqrt{x^2 + 2x + 7}}$$

$$ឆ. \lim_{x \rightarrow -\infty} \sqrt[3]{\frac{2x - 1}{7 - 16x}}$$

$$ឈ. \lim_{x \rightarrow +\infty} \left(\frac{x}{3x + 1} \right) \left(\frac{4x^2 + 1}{2x^2 + x} \right)^3$$

$$ខ. \lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 1} - 3}{3x - 5}$$

$$ឃ. \lim_{x \rightarrow +\infty} \frac{\sqrt{8x^2 - 1} + x}{\sqrt{2x^2 - 1} + 2x}$$

$$ច. \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 9} - \sqrt{x^2 - 9}}{6x}$$

$$ជ. \lim_{x \rightarrow -\infty} \frac{1 + 7\sqrt[3]{x}}{2\sqrt[3]{x}}$$

$$ញ. \lim_{x \rightarrow +\infty} \ln \left(\frac{x}{x + 8} \right)$$

• លីមីតរាងមិនកំណត់ $+\infty - \infty$

ដើម្បីគណនាលីមីតរាងមិនកំណត់ $+\infty - \infty$ គេត្រូវអនុវត្តតាមវិធានដូចខាងក្រោម៖

ចំពោះអនុគមន៍ពហុធា៖

- ទាញយកតួដ៏ក្រៃខ្ពស់ជាងគេ ទាំងភាគយក និងភាគបែង
- រួចគណនាលីមីតនៃប្រភាគថ្មី ។

ចំពោះអនុគមន៍អសនិទាន៖

- គុណ និងចែកកន្សោមនោះជាមួយកន្សោមឆ្លាស់ ឬកន្សោមបំពេញរបស់វា
- រួចគណនាលីមីតនៃប្រភាគថ្មី ។

អនុវត្ត ៤៖

12. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\text{ក. } \lim_{x \rightarrow +\infty} (2x^4 - x^2 + 1)$$

$$\text{ខ. } \lim_{x \rightarrow +\infty} (2x^3 + x^2 - 3x + 1)$$

$$\text{គ. } \lim_{x \rightarrow +\infty} (x^4 - 2x^3 + x - 2)$$

$$\text{ឃ. } \lim_{x \rightarrow -\infty} (-x^3 + x^2 - 4x + 5)$$

$$\text{ង. } \lim_{x \rightarrow +\infty} (7x^2 - 3x + 1)$$

$$\text{ច. } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 1} - x)$$

$$\text{ឆ. } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 2} - x + 1)$$

$$\text{ជ. } \lim_{x \rightarrow +\infty} (\sqrt{x^2 + x} - x^2)$$

$$\text{ល. } \lim_{x \rightarrow +\infty} (\sqrt{4x^2 + x + 1} + 2x)$$

$$\text{ញ. } \lim_{x \rightarrow +\infty} (\sqrt{x^2 - 4x} - x + 2023)$$

$$\text{ដ. } \lim_{x \rightarrow +\infty} (2x - 3 - \sqrt{4x^2 + 4x + 3})$$

$$\text{ប. } \lim_{x \rightarrow +\infty} (x - 3 - \sqrt{x^2 - x + 1})$$

$$\text{ខ. } \lim_{x \rightarrow +\infty} (x^2 - \sqrt{x^4 + 7x^2 + 1})$$

$$\text{ឈ.}$$

13. កំណត់តម្លៃនៃចំនួនថេរ a, p, q, m, n ដើម្បីឲ្យលីមីតខាងក្រោមជាចំនួនថេរ រួចគណនាលីមីតនោះ៖

$$\text{ក. } \lim_{x \rightarrow 1} \frac{x^2 - ax + 1}{x - 1}$$

$$\text{ខ. } \lim_{x \rightarrow 2} \frac{\sqrt{ax + 1} - 3}{x - 2}$$

$$\text{គ. } \lim_{x \rightarrow +\infty} \frac{bx^2 - x}{x - 2}$$

$$\text{ឃ. } \lim_{x \rightarrow 4} \frac{\sqrt{5x - a} - 4}{x^2 - 16}$$

14. កំណត់តម្លៃនៃចំនួនថេរ a, b ដើម្បីឲ្យ៖

$$\text{ក. } \lim_{x \rightarrow 1} \frac{x^2 + ax + b}{x - 1} = 5$$

$$\text{ខ. } \lim_{x \rightarrow 5} \frac{\sqrt{x - a} - \sqrt{b - 5}}{x^2 - 25} = \frac{1}{40}$$

$$\text{គ. } \lim_{x \rightarrow \pm\infty} \frac{ax^2 + bx + 8}{2x - 3} = 3$$

$$\text{ឃ. } \lim_{x \rightarrow -2} \frac{x^2 + ax - 6}{2x^2 + 3x - 2} = b$$

$$\text{ង. } \lim_{x \rightarrow 2} \frac{x^2 - mx + 8}{x^2 - (n + 2)x + 2n} = \frac{1}{5}$$

$$\text{ច. } \lim_{x \rightarrow +\infty} \frac{(a - 1)x^2 + bx + 1}{2x + 3} = 1$$

15. កំណត់អនុគមន៍ដ៏ក្រេទីបនៃ $f(x)$ ដើម្បីឲ្យវាបំពេញលក្ខខណ្ឌទាំងពីរខាងក្រោម៖

$$\begin{cases} \lim_{x \rightarrow +\infty} \frac{f(x)}{x^2 - 1} = 2 \\ \lim_{x \rightarrow 1} \frac{f(x)}{x^2 - 1} = -1 \end{cases}$$

៤. លីមីតនៃអនុគមន៍ត្រីកោណមាត្រ

រូបមន្តត្រីកោណមាត្រសំខាន់ៗ

$$1. \lim_{x \rightarrow 0} \frac{\sin ax}{ax} = \lim_{x \rightarrow 0} \frac{ax}{\sin ax} = 1$$

$$2. \lim_{x \rightarrow 0} \frac{\tan ax}{ax} = \lim_{x \rightarrow 0} \frac{ax}{\tan ax} = 1$$

$$3. \lim_{x \rightarrow 0} \frac{1 - \cos ax}{ax} = 0$$

$$4. \lim_{x \rightarrow 0} \frac{1 - \cos kx}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{kx}{2} \right)}{x^2} = \frac{k^2}{2}$$

$$* 1 - \cos ax = 2 \sin^2 \frac{ax}{2}$$

$$* \sin 2x = 2 \sin x \cos x$$

$$* \sin^2 x + \cos^2 x = 1$$

$$* 1 + \cos ax = 2 \cos^2 \frac{ax}{2}$$

$$* 1 + \tan^2 x = \frac{1}{\cos^2 x}$$

$$* 1 + \cot^2 x = \frac{1}{\sin^2 x}$$

លំហាត់គំរូ

$$ក. \lim_{x \rightarrow 0} \frac{\sin 2023x}{x} = \lim_{x \rightarrow 0} \frac{\sin 2023x}{2023x} \times 2023 = 1 \times 2023 = \boxed{2023}$$

$$ខ. \lim_{x \rightarrow 0} \frac{4x}{\tan 5x} = \lim_{x \rightarrow 0} \frac{5x}{\tan 5x} \times \frac{4}{5} = 1 \times \frac{4}{5} = \boxed{\frac{4}{5}}$$

$$គ. \lim_{x \rightarrow 0} \frac{1 - \cos 3x}{2x^2} \text{ រាងមិនកំណត់ } \frac{0}{0}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin ax}{ax} &= 1 \\ \lim_{x \rightarrow 0} \frac{\sin^2(ax)}{(ax)^2} &= 1 \\ \lim_{x \rightarrow 0} \frac{\sin^3(ax)}{(ax)^3} &= 1 \end{aligned}$$

គេបាន: $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{2x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{3x}{2} \right)}{2x^2}$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 \left(\frac{3x}{2} \right)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 \left(\frac{3x}{2} \right)}{\left(\frac{3x}{2} \right)^2} \times \left(\frac{3}{2} \right)^2 = 1 \times \frac{9}{4} = \boxed{\frac{9}{4}}$$

ឃ. $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 6x}{x^2}$ រាងមិនកំណត់ $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos x \cos 6x}{x^2} &= \lim_{x \rightarrow 0} \frac{1 - \cancel{\cos x} + \cancel{\cos x} - \cos x \cos 6x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} + \frac{\cos x (1 - \cos 6x)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{x}{2} \right)}{x^2} + \frac{\cos x \times 2 \sin^2 (3x)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{2 \sin^2 \left(\frac{x}{2} \right)}{\left(\frac{x}{2} \right)^2 \times 4} + \frac{2 \sin^2 (3x)}{(3x)^2} \times 9 \\ &= \frac{2}{4} + 2 \times 9 = \frac{1}{2} + 18 = \boxed{\frac{37}{2}} \end{aligned}$$

អនុវត្ត ៥៖

16. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម:

ក. $\lim_{x \rightarrow 0} \frac{\sin 12x}{-x}$

ខ. $\lim_{x \rightarrow 0} \frac{x + \sin 3x}{x}$

គ. $\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan(-6x)}$

ឃ. $\lim_{x \rightarrow 0} \frac{\sin^3 3x + 5x^3}{x^3}$

ង. $\lim_{x \rightarrow 0} \frac{2 \sin x + \sin 2x}{x}$

ច. $\lim_{x \rightarrow 0} \frac{\sin 2x \times \sin 3x}{5x^2}$

ឆ. $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \sin 2x}$

ជ. $\lim_{x \rightarrow 0} \frac{2 \sin 3x}{2x - 3 \sin 2x}$

ឈ. $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{4x}$

ញ. $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$

ដ. $\lim_{x \rightarrow 0} \frac{-6x}{\sin 2x}$

ប. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\tan^2 x}$

ខ. $\lim_{x \rightarrow 0} \frac{\tan x + \sin x}{x \cos x}$

ឈ. $\lim_{x \rightarrow 0} \frac{\sin 2x}{2x^2 + x}$

ណ. $\lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x \cos x}$

ក. $\lim_{x \rightarrow 0} \frac{\sin^2 x \cos x}{1 - \cos x}$

17. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 0} \frac{-2 \sin 2x}{1 - \sqrt{x+1}}$

ខ. $\lim_{x \rightarrow 0} \frac{-2 \sin 5x}{\sqrt{5} - \sqrt{x+5}}$

គ. $\lim_{x \rightarrow 0} \frac{\sin 7x}{\sqrt{x+9} - 3}$

ច. $\lim_{x \rightarrow 0} \frac{\sin 3x}{2(2 - \sqrt{4+x})}$

ឃ. $\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{\sin 5x}$

ង. $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2}$

ឆ. $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sin 5x}$

ជ. $\lim_{x \rightarrow 0} \frac{\sqrt{2x+3} - \sqrt{3}}{\sin 3x}$

18. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sin 3x}$

ខ. $\lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$

គ. $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$

ឃ. $\lim_{x \rightarrow 0} \frac{1+x-\cos 2x}{1-\sqrt{1+\sin x}}$

ង. $\lim_{x \rightarrow 0} \frac{1-\cos x \sqrt{\cos 2x}}{x^2}$

ច. $\lim_{x \rightarrow 0} \frac{\sqrt{1-\tan x} - \sqrt{1+\tan x}}{2x}$

19. ដោយប្រើ $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ ។ ចូរគណនាលីមីតខាងក្រោម:

ក. $\lim_{x \rightarrow 0} \frac{2x}{\sin x}$

ខ. $\lim_{x \rightarrow 0} \frac{1-\cos^2 x}{\sin^2 x}$

គ. $\lim_{x \rightarrow 0} \frac{8x^2 - \sin x}{x}$

លីមីតនៃអនុគមន៍ត្រីកោណមាត្រ កាលណា $x \rightarrow x_0$, $x_0 \neq 0$

ដើម្បីគណនាលីមីតរវាងមិនកំណត់នៃអនុគមន៍ត្រីកោណមាត្រ កាលណា $x \rightarrow x_0$ គេត្រូវ:

- តាង $t = x - x_0$ ឬ $t = x_0 - x$ កាលណា $x \rightarrow x_0$ នោះ $t \rightarrow 0$
- ជំនួស x ដោយ $t + x_0$ ឬ $x_0 - t$ ក្នុងលីមីត រួចប្រើរូបមន្តត្រីកោណមាត្រសំខាន់ៗខាងលើ ។

លំហាត់គំរូ

គណនាលីមីតនៃអនុគមន៍ខាងក្រោម:

ក. $\lim_{x \rightarrow 4} \frac{\sin 4\pi x}{x-4}$ រវាងមិនកំណត់ $\frac{0}{0}$

តាង $t = x - 4 \Rightarrow x = t + 4$ បើ $x \rightarrow 4$ នោះ $t \rightarrow 0$

គេបាន: $\lim_{t \rightarrow 0} \frac{\sin 4\pi(t+4)}{t+4-4} = \lim_{t \rightarrow 0} \frac{\sin(4\pi t + 16\pi)}{t}$
 $= \lim_{t \rightarrow 0} \frac{\sin 4\pi t}{t}$
 $= \lim_{t \rightarrow 0} \frac{\sin 4\pi t}{4\pi t} \times 4\pi$
 $= 1 \times 4\pi = \boxed{4\pi}$

$\sin(2k\pi + \alpha) = \sin \alpha$
 $\sin(2k\pi - \alpha) = -\sin \alpha$

ខ. $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\sin\left(\frac{\pi}{6} - x\right)}{2x - \frac{\pi}{3}}$ រាងមិនកំណត់ $\frac{0}{0}$

តាង $t = x - \frac{\pi}{6} \Rightarrow x = t + \frac{\pi}{6}$

បើ $x \rightarrow \frac{\pi}{6}$ នោះ $t \rightarrow 0$

គេបាន: $\lim_{t \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} - \left(t + \frac{\pi}{6}\right)\right)}{2\left(t + \frac{\pi}{6}\right) - \frac{\pi}{3}} = \lim_{t \rightarrow 0} \frac{\sin(-t)}{2t + \frac{\pi}{3} - \frac{\pi}{3}}$

$$= \lim_{t \rightarrow 0} \frac{-\sin t}{2t} = \boxed{-\frac{1}{2}}$$

$$\sin(-x) = -\sin x$$

$$\cos(-x) = \cos x$$

គ. $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\cos x - \sqrt{3} \sin x}{3x - \frac{\pi}{2}}$ រាងមិនកំណត់ $\frac{0}{0}$

$$\begin{aligned} \cos x - \sqrt{3} \sin x &= 2 \left(\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x \right) \\ &= 2 \left(\sin \frac{\pi}{6} \cos x - \cos \frac{\pi}{6} \sin x \right) \\ &= 2 \left(\sin \left(\frac{\pi}{6} - x \right) \right) \end{aligned}$$

$$.... a = 1, b = -\sqrt{3}$$

$$r = \sqrt{a^2 + b^2} = \sqrt{1 + (-\sqrt{3})^2}$$

$$r = \sqrt{1 + 3} = \sqrt{4} = 2$$

$$\sin(a - b) = \sin a \cos b - \sin b \cos a$$

គេបាន: $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\cos x - \sqrt{3} \sin x}{3x - \frac{\pi}{2}} = \lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin\left(\frac{\pi}{6} - x\right)}{3x - \frac{\pi}{2}}$

តាង $t = \frac{\pi}{6} - x \Rightarrow x = \frac{\pi}{6} - t$ បើ $x \rightarrow \frac{\pi}{6}$ នោះ $t \rightarrow 0$

គេបាន: $\lim_{t \rightarrow 0} \frac{2 \sin t}{3\left(\frac{\pi}{6} - t\right) - \frac{\pi}{2}} = \lim_{t \rightarrow 0} \frac{2 \sin t}{\frac{\pi}{2} - 3t - \frac{\pi}{2}} = \lim_{t \rightarrow 0} \frac{2 \sin t}{-3t} = \boxed{-\frac{2}{3}}$

ឃ. $\lim_{x \rightarrow 1} \frac{1-x}{\sin \frac{2\pi}{x}}$ រាងមិនកំណត់ $\frac{0}{0}$

ករណី $\sin \frac{a}{x}$ គេត្រូវតាង $t = \frac{1}{x}$

តាង $t = \frac{1}{x} \Rightarrow x = \frac{1}{t}$ បើ $x \rightarrow 1$ នោះ $t \rightarrow \frac{1}{2}$

គេបាន: $\lim_{x \rightarrow 1} \frac{1-x}{\sin \frac{2\pi}{x}} = \lim_{t \rightarrow \frac{1}{2}} \frac{1-\frac{1}{t}}{\sin 2\pi t} = \lim_{t \rightarrow \frac{1}{2}} \frac{t-1}{t \sin 2\pi t}$

តាង $u = t-1 \Rightarrow t = u+1$ បើ $t \rightarrow \frac{1}{2}$ នោះ $u \rightarrow 0$

គេបាន:

$$\begin{aligned} \lim_{t \rightarrow \frac{1}{2}} \frac{t-1}{t \sin 2\pi t} &= \lim_{u \rightarrow 0} \frac{u}{(u+1) \sin [2\pi(u+1)]} \\ &= \lim_{u \rightarrow 0} \frac{u}{(u+1) \sin (2\pi u + 2\pi)} \\ &= \lim_{u \rightarrow 0} \frac{u}{(u+1) \sin 2\pi u} \\ &= \lim_{u \rightarrow 0} \frac{2\pi u}{\sin 2\pi u} \times \frac{1}{2\pi(u+1)} \\ &= \boxed{\frac{1}{2\pi}} \end{aligned}$$

ង. $\lim_{x \rightarrow 2} \frac{2-x}{\sin \frac{2\pi}{x}}$ រាង $\frac{0}{0}$

តាង $t = \frac{1}{x} \Rightarrow x = \frac{1}{t}$ បើ $x \rightarrow 2$ នោះ $t \rightarrow \frac{1}{2}$

គេបាន: $\lim_{x \rightarrow 2} \frac{2-x}{\sin \frac{2\pi}{x}} = \lim_{t \rightarrow \frac{1}{2}} \frac{2-\frac{1}{t}}{\sin 2\pi t} = \lim_{t \rightarrow \frac{1}{2}} \frac{2t-1}{t \sin 2\pi t}$

តាង $u = t - \frac{1}{2} \Rightarrow t = u + \frac{1}{2}$ បើ $t \rightarrow \frac{1}{2}$ នោះ $u \rightarrow 0$

$$\begin{aligned} \text{គេបាន: } \lim_{t \rightarrow \frac{1}{2}} \frac{2t-1}{t \sin 2\pi t} &= \lim_{u \rightarrow 0} \frac{2\left(u + \frac{1}{2}\right) - 1}{\left(u + \frac{1}{2}\right) \sin 2\pi \left(u + \frac{1}{2}\right)} \\ &= \lim_{u \rightarrow 0} \frac{2u}{\left(u + \frac{1}{2}\right) \sin(2\pi u + \pi)} \\ &= \lim_{u \rightarrow 0} \frac{2u}{\left(u + \frac{1}{2}\right) (-\sin 2\pi u)} \\ &= \lim_{u \rightarrow 0} \frac{2\pi u}{\sin 2\pi u} \times \frac{1}{\pi} \times \frac{1}{-\left(u + \frac{1}{2}\right)} \\ &= \frac{1}{\pi} \times \left(\frac{1}{-\frac{1}{2}} \right) = \boxed{-\frac{2}{\pi}} \end{aligned}$$

អនុវត្ត ៦៖

20. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

ក. $\lim_{x \rightarrow 3} \frac{\sin 4\pi x}{3 - x}$

ខ. $\lim_{x \rightarrow 1} \frac{1 + \cos \pi x}{(x - 1)^2}$

គ. $\lim_{x \rightarrow 2} \frac{\sin \pi x}{2 - x}$

ឃ. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin 4x}{4x - \pi}$

ង. $\lim_{x \rightarrow 1} \frac{\cos \frac{\pi}{2} x}{1 - x}$

ច. $\lim_{x \rightarrow 1} \frac{x^2 - 1 + \sin \pi x}{x - 1}$

$$\text{ឆ. } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin\left(x - \frac{\pi}{3}\right)}{\frac{\pi}{3} - x}$$

$$\text{ជ. } \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 - \sqrt{1 + \cos x}}{\pi - 2x} \right)$$

$$\text{ឈ. } \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin\left(x - \frac{\pi}{4}\right)}{\left(\frac{\pi}{4} - x\right)}$$

$$\text{ញ. } \lim_{x \rightarrow \pi} (\pi - x) \tan \frac{x}{2}$$

$$\text{ដ. } \lim_{x \rightarrow \frac{\pi}{2}} \frac{\pi - 2x}{\cos x}$$

$$\text{ប. } \lim_{x \rightarrow 2} \frac{2 - x}{\sin \frac{2\pi}{x}}$$

$$\text{ខ. } \lim_{x \rightarrow \pi} \frac{\cos 2x - 1}{x - \pi}$$

$$\text{ឈ. } \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\pi - 4x}$$

$$\text{ណ. } \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin x - \sqrt{3} \cos x}{3x - \pi}$$

$$\text{ក. } \lim_{x \rightarrow 4} \frac{1 - \cos \pi x}{(4 - x)^2}$$

$$\text{ច. } \lim_{x \rightarrow 2} \frac{x^3 - 8}{\cos \frac{\pi}{x}}$$

$$\text{ទ. } \lim_{x \rightarrow 2} \frac{\sin(x - 2)}{2x^2 - 8x + 8}$$

$$\text{ត. } \lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin x - 1}{4 \cos^2 x - 3}$$

$$\text{ស. } \lim_{x \rightarrow \frac{\pi}{2}} (\pi - 2x) \tan x$$

$$\text{ហ. } \lim_{x \rightarrow \infty} (3x + 1) \tan \frac{2}{x}$$

$$\text{ជ. } \lim_{x \rightarrow 2} (\pi^2 - x^2) \tan \frac{x}{2}$$

$$\text{ណ. } \lim_{x \rightarrow 1} \frac{\sin(x - 1)}{2x - 2}$$

$$\text{ក. } \lim_{x \rightarrow 2\pi} \frac{x - 2\pi}{\sin x}$$

$$21. \text{ គេឱ្យ } f(x) = \sin x \text{ ចូរគណនាលីមីត } \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{4} + h\right) - f\left(\frac{\pi}{4}\right)}{h} \text{ ។}$$

$$22. \text{ គេឱ្យ } f(x) = \cos x \text{ ចូរគណនាលីមីត } \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{6} + h\right) - f\left(\frac{\pi}{6}\right)}{h}$$

៥. លីមីតនៃអនុគមន៍អិចស្ប៉ូណង់ស្យែល

រូបមន្តគ្រឹះសំខាន់ៗ

1. $\lim_{x \rightarrow +\infty} e^x = +\infty$	2. $\lim_{x \rightarrow -\infty} e^x = 0$	3. $\lim_{x \rightarrow +\infty} \frac{x^n}{e^x} = 0, (n > 0)$
4. $\lim_{x \rightarrow +\infty} \frac{e^x}{x^n} = +\infty, (n > 0)$	5. $\lim_{x \rightarrow -\infty} x e^x = 0$	6. $\lim_{x \rightarrow +\infty} x^n e^x = +\infty, (n > 0)$
7. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$	8. $\lim_{x \rightarrow 0} \frac{e^u - 1}{u} = 1$	9. $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$
10. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$	11. $\lim_{u \rightarrow 0} (1+u)^{\frac{1}{u}} = e$ 12. $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$ $\lim_{u \rightarrow +\infty} \left(1 + \frac{1}{u}\right)^u = e$ $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e$	

លក្ខណៈ: ក. $e^x \times e^y = e^{x+y}$ ខ. $\frac{e^x}{e^y} = e^{x-y}$ គ. $e^0 = 1$ ឃ. $(e^x)^y = e^{xy}$

លំហាត់គ្រឹះ: គណនាលីមីតខាងក្រោម៖

ក. $\lim_{x \rightarrow \pm\infty} (x^3 - x + 3)e^{-x}$

បើ $x \rightarrow +\infty$ គេបាន: $\lim_{x \rightarrow +\infty} (x^3 - x + 3)e^{-x} = \boxed{0}$ ព្រោះ: $\lim_{x \rightarrow +\infty} e^{-x} = 0$

បើ $x \rightarrow -\infty$ គេបាន: $\lim_{x \rightarrow -\infty} (x^3 - x + 3)e^{-x}$
 $= \lim_{x \rightarrow -\infty} e^{-x} x^3 \left(1 - \frac{1}{x^2} + \frac{3}{x^3}\right)$
 $= \lim_{x \rightarrow -\infty} e^{-x} x^3 = e^{+\infty} (-\infty)^3 = +\infty (-\infty) = \boxed{-\infty}$

ខ. $\lim_{x \rightarrow +\infty} \left(1 + \frac{10}{x}\right)^x$ រាងមិនកំណត់ 1^∞

គេបាន: $\lim_{x \rightarrow +\infty} \left(1 + \frac{10}{x}\right)^x = \lim_{x \rightarrow +\infty} \left(1 + \frac{10}{x}\right)^{\frac{x}{10} \times 10}$

$$= \left[\lim_{x \rightarrow +\infty} \left(1 + \frac{10}{x}\right)^{\frac{x}{10}} \right]^{10}$$

$$= e^{10}$$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{a}{x}\right)^{\frac{x}{a}} = e$$

គ. $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}}$ រាងមិនកំណត់ 1^∞

$$\lim_{x \rightarrow 0} (1 + ax)^{\frac{1}{ax}} = e$$

គេបាន: $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}} = \lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{2x} \times 2}$

$$= \lim_{x \rightarrow 0} \left[(1 + 2x)^{\frac{1}{2x}} \right]^2 = \boxed{e^2}$$

ឃ. $\lim_{x \rightarrow 0} \frac{e^{3 \sin x} - e^{\sin 3x}}{\sin^3 2x}$ រាងមិនកំណត់ $\frac{0}{0}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{e^{3 \sin x} - e^{\sin 3x}}{\sin^3 2x} &= \lim_{x \rightarrow 0} \frac{e^{3 \sin x} - e^{3 \sin x - 4 \sin^3 x}}{\sin^3 2x} = \lim_{x \rightarrow 0} \frac{e^{3 \sin x} (1 - e^{-4 \sin^3 x})}{\sin^3 2x} \\ &= \lim_{x \rightarrow 0} \frac{-e^{3 \sin x} (e^{-4 \sin^3 x} - 1)}{\sin^3 2x} = \lim_{x \rightarrow 0} \frac{-e^{\sin 3x} \left(\frac{e^{-4 \sin^3 x} - 1}{x^3} \right)}{\left(\frac{\sin^3 2x}{x^3} \right)} \\ &= \lim_{x \rightarrow 0} \frac{-e^{\sin 3x} \left(\frac{e^{-4 \sin^3 x} - 1}{-4 \sin^3 x} \times \frac{-4 \sin^3 x}{x^3} \right)}{\left(\frac{\sin^3 2x}{(2x)^3} \times 2^3 \right)} \\ &= - \frac{e^0 \times 1 \times (-4)}{1 \times 2^3} = \frac{4}{8} = \frac{1}{2} \end{aligned}$$

អនុវត្ត ល៖

23. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\begin{array}{lll} \text{ក. } \lim_{x \rightarrow \pm\infty} (x^3 - x + 3)e^{-x} & \text{ខ. } \lim_{x \rightarrow \pm\infty} (x^2 + xe^x) & \text{គ. } \lim_{x \rightarrow \pm\infty} (2 + e^x) \\ \text{ឃ. } \lim_{x \rightarrow -\infty} (202e^x + 203) & \text{ង. } \lim_{x \rightarrow -\infty} (x - 2 + e^{1-x}) & \text{ច. } \lim_{x \rightarrow +\infty} (x^3 + x)e^{-x} \\ \text{ឆ. } \lim_{x \rightarrow +\infty} (e^x - 3x + 5) & \text{ជ. } \lim_{x \rightarrow \pm\infty} (2 - x)(e^x + 1) & \text{ឈ. } \lim_{x \rightarrow +\infty} (x^3 - 2x)e^{-x} \\ \text{ញ. } \lim_{x \rightarrow \pm\infty} (e^{2x} - 2x - 1) & \text{ដ. } \lim_{x \rightarrow \pm\infty} (x - 2 + xe^{2x}) & \text{ប. } \lim_{x \rightarrow \pm\infty} (x^3 e^x - x + 1) \end{array}$$

24. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\begin{array}{lll} \text{ក. } \lim_{x \rightarrow +\infty} \frac{e^x - x}{2e^x + 1} & \text{ខ. } \lim_{x \rightarrow -\infty} \frac{e^x - x}{2e^x + 1} & \text{គ. } \lim_{x \rightarrow +\infty} \frac{x^2 - x + 1}{(e^x + 2)(e^x - 1)} \\ \text{ឃ. } \lim_{x \rightarrow \pm\infty} \frac{3e^x + 1}{5e^x - 2} & \text{ង. } \lim_{x \rightarrow \pm\infty} \frac{2x + 1}{x + xe^{-x}} & \text{ច. } \lim_{x \rightarrow \pm\infty} \frac{2e^x - x}{1 + e^x} \\ \text{ឆ. } \lim_{x \rightarrow \pm\infty} \frac{xe^{2x} + 1}{e^{2x} - 3} & \text{ជ. } \lim_{x \rightarrow \pm\infty} \frac{e^{2x} - x^2}{3e^x - 3x} & \text{ឈ. } \lim_{x \rightarrow \pm\infty} \frac{xe^{2x} - x}{e^{2x} - 2e^x + 1} \\ \text{ញ. } \lim_{x \rightarrow \pm\infty} \frac{e^x - 1}{2x + 1} & \text{ដ. } \lim_{x \rightarrow +\infty} \frac{2x - 1}{e^x + x} & \text{ប. } \lim_{x \rightarrow \pm\infty} \frac{2e^{2x} - e^x + 1}{e^{2x} + 3e^x + 3} \end{array}$$

25. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\begin{array}{lll} \text{ក. } \lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x} & \text{ខ. } \lim_{x \rightarrow 0} \frac{e^{2023x} - 1}{2024x} & \text{គ. } \lim_{x \rightarrow 0} \frac{e^{-6x} - 1}{x} \\ \text{ឃ. } \lim_{x \rightarrow 0} \frac{\sin 2x}{1 - e^x} & \text{ង. } \lim_{x \rightarrow 0} \frac{e^{3x} - 1}{1 - e^{2x}} & \text{ច. } \lim_{x \rightarrow 0} \frac{e^{2x} + \sin 2x - 1}{x} \\ \text{ឆ. } \lim_{x \rightarrow 0} \frac{(e^{2+x} - e^2) \sin \pi x}{x^2} & \text{ជ. } \lim_{x \rightarrow 0} \frac{e^{ax} - x^{bx}}{x} & \text{ឈ. } \lim_{x \rightarrow 0} \frac{1 - e^x - \sin x}{1 - \sqrt{x + 1}} \\ \text{ញ. } \lim_{x \rightarrow 0} \frac{1 - e^{2023x}}{\sin 2012x} & \text{ដ. } \lim_{x \rightarrow 0} \frac{e^{2x} - 2e^x + 1}{x^2} & \text{ប. } \lim_{x \rightarrow 0} \frac{1 - \cos^4 x}{e^{\sin^2 x} - 1} \\ \text{ខ. } \lim_{x \rightarrow 1} \frac{e^{2x-1} - e^x + \sin \pi x}{x^2 - 1} & \text{ឈ. } \lim_{x \rightarrow 0} \frac{2e^{2x} + 3e^x - 5}{e^x - 1} & \text{ណ. } \lim_{x \rightarrow 0} \frac{2e^{2x} + 3e^{3x} - 5}{x} \end{array}$$

26. គណនាលីមីត រាងមិនកំណត់ 1^∞ ៖

$$\text{ក. } \lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}}$$

$$\text{ខ. } \lim_{x \rightarrow +\infty} \left(1 + \frac{2}{x}\right)^x$$

$$\text{គ. } \lim_{x \rightarrow 0} (1 + x)^{\frac{3}{x}}$$

$$\text{ឃ. } \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^{\frac{x^2+1}{x}}$$

$$\text{ង. } \lim_{x \rightarrow 1} (2 - x)^{\frac{-3}{x^2-1}}$$

$$\text{ច. } \lim_{x \rightarrow +\infty} \left(\frac{x+3}{x-1}\right)^{x+3}$$

$$\text{ឆ. } \lim_{x \rightarrow +\infty} \left(\frac{x^2+1}{x^2-1}\right)^{x^2}$$

$$\text{ជ. } \lim_{x \rightarrow +\infty} \left(\frac{x+1}{x-2}\right)^{2x-1}$$

$$\text{ណ. } \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$$

$$\text{ញ. } \lim_{x \rightarrow 1} (-2x + 3)^{\frac{x+3}{x(x-1)}}$$

$$\text{ដ. } \lim_{x \rightarrow 0} (3 - 2e^x)^{\frac{3x-1}{x}}$$

$$\text{ប. } \lim_{x \rightarrow 1} (2 - x)^{\frac{-3}{x^2-1}}$$

៥. លីមីតនៃអនុគមន៍លោការីត

រូបមន្តគ្រឹះសំខាន់ៗ

$$1. \lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$2. \lim_{x \rightarrow +\infty} \ln x = +\infty$$

$$3. \lim_{x \rightarrow +\infty} \frac{\ln x}{x^n} = 0, (n > 0)$$

$$4. \lim_{x \rightarrow +\infty} \frac{x^n}{\ln x} = +\infty, (n > 0)$$

$$5. \lim_{x \rightarrow 0^+} x^n \ln x = 0, (n > 0)$$

$$6. \lim_{x \rightarrow 0} \frac{\ln(1 + ax)}{ax} = 1$$

$$7. \lim_{x \rightarrow 0} \frac{\ln(1 + u)}{u} = 1$$

$$8. \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0$$

$$9. \lim_{x \rightarrow +\infty} \frac{x}{\ln x} = +\infty$$

$$\text{លក្ខណៈ: ក. } \ln 1 = 0$$

$$\text{ខ. } \ln e = 1$$

$$\text{គ. } \ln e^a = a$$

$$\text{ឃ. } e^{\ln a} = a$$

$$\text{ង. } e^{b \ln a} = a^b$$

$$\text{ច. } \ln(ab) = \ln a + \ln b$$

$$\text{ឆ. } \ln \frac{a}{b} = \ln a - \ln b$$

លំហាត់គំរូ

គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$\text{ក. } \lim_{x \rightarrow 0^+} (2x - 4 + \ln x) = 2 \times 0^+ - 4 + \ln 0^+ = \boxed{-\infty} \quad \text{ព្រោះ: } \lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$\begin{aligned} \text{ខ. } \lim_{x \rightarrow +\infty} (2x^2 - 3 \ln x) &= \lim_{x \rightarrow +\infty} x^2 \left(2 - \frac{3 \ln x}{x^2}\right) \\ &= \lim_{x \rightarrow +\infty} 2x^2 = \boxed{+\infty} \end{aligned}$$

ពេល $x \rightarrow +\infty$, $x^2 > \ln x$

ហើយ $\frac{\ln x}{x^2} \rightarrow 0$

$$\begin{aligned} \text{គ. } \lim_{x \rightarrow 0} \frac{\ln(1 + 2023x)}{x} &= \lim_{x \rightarrow 0} \frac{\ln(1 + 2023x)}{2023x} \times 2023 \\ &= 1 \times 2023 = \boxed{2023} \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\ln(1 + kx)}{kx} = 1$$

$$\text{ឃ. } \lim_{x \rightarrow 0^+} \frac{1}{x} [\ln(x+2) - \ln(x-1)]$$

គេបាន:

$$\lim_{x \rightarrow +\infty} \frac{1}{x} [\ln(x+2) - \ln(x-1)] = \lim_{x \rightarrow +\infty} \frac{1}{x} \ln\left(\frac{x+2}{x-1}\right)$$

ប្រើរូបមន្ត:

$$\ln a - \ln b = \ln \frac{a}{b}$$

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{ax}\right)^{ax} = e$$

$$\lim_{x \rightarrow +\infty} \ln e^{ax} = \ln \lim_{x \rightarrow +\infty} e^{ax}$$

$$= \lim_{x \rightarrow +\infty} \ln \left(1 - 1 + \frac{x+2}{x-1}\right)^{\frac{1}{x}}$$

$$= \ln \lim_{x \rightarrow +\infty} \left(1 + \frac{x+2-x+1}{x-1}\right)^{\frac{1}{x}}$$

$$= \ln \lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x-1}\right)^{\frac{1}{x}}$$

$$= \ln \lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x-1}\right)^{\frac{x-1}{3} \times \frac{3}{x-1} \times \frac{1}{x}}$$

$$= \ln \lim_{x \rightarrow +\infty} e^{\frac{3}{(x-1)x}} = \ln e^0 = \ln 1 = \boxed{0}$$

អនុវត្ត ៨:

27. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម:

$$\text{ក. } \lim_{x \rightarrow 0^+} (x - 3 - \ln x)$$

$$\text{ខ. } \lim_{x \rightarrow +\infty} (x - 3 - \ln x)$$

$$\text{គ. } \lim_{x \rightarrow 0^+} (3x - x^2 \ln x)$$

$$\text{ឃ. } \lim_{x \rightarrow 4^+} x \ln(4 - 3x - x^2)$$

$$\text{ង. } \lim_{x \rightarrow 1^-} x \ln(4 - 3x - x^2)$$

$$\text{ច. } \lim_{x \rightarrow +\infty} \frac{\ln x - 2}{x^3}$$

$$\text{ឆ. } \lim_{x \rightarrow 0^+} \frac{\ln x + 5}{2x^2 - 3}$$

$$\text{ជ. } \lim_{x \rightarrow +\infty} \ln\left(\frac{3x+5}{x+2}\right)$$

$$\text{ឈ. } \lim_{x \rightarrow 0^+} \frac{x^2 + 1}{\ln x + 2}$$

$$\text{ញ. } \lim_{x \rightarrow +\infty} \ln\left(\frac{2x+1}{x+1}\right)$$

$$\text{ដ. } \lim_{x \rightarrow +\infty} \frac{\ln(x^2 + 4)}{\ln(2x + 1)}$$

$$\text{ប. } \lim_{x \rightarrow +\infty} \frac{e^x \ln x + 1}{x^2}$$

$$ខ. \lim_{x \rightarrow +\infty} \frac{e^x + \ln x}{x^3}$$

$$ឆ. \lim_{x \rightarrow 0^+} \frac{5 - 3 \ln x}{\ln x + 2}$$

$$ណ. \lim_{x \rightarrow +\infty} \frac{\ln^2 x + 5}{2x^2 - 3}$$

28. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow +\infty} [\ln(2x+1) - \ln(x+2)]$$

$$ខ. \lim_{x \rightarrow 0^+} (x^2 - 3x + \ln x)$$

$$គ. \lim_{x \rightarrow +\infty} [\ln(e^5 x + 1) - \ln(x+2)]$$

$$ឃ. \lim_{x \rightarrow +\infty} x [\ln(x+1) - \ln x]$$

29. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow +\infty} \frac{e^x + \ln x}{x^3}$$

$$ខ. \lim_{x \rightarrow 0^+} \frac{5 - 3 \ln x}{\ln x + 2}$$

$$គ. \lim_{x \rightarrow +\infty} \frac{\ln^2 x + 5}{2x^2 - 3}$$

$$ឃ. \lim_{x \rightarrow +\infty} \frac{\ln(2 + e^{3x})}{\ln(1 + e^{2x})}$$

$$ង. \lim_{x \rightarrow 0^+} \frac{x^2 - 1}{x^3 + \ln x}$$

$$ច. \lim_{x \rightarrow 0^+} \frac{e^x \ln x + 1}{x^2}$$

30. គណនាលីមីតនៃអនុគមន៍ខាងក្រោម៖

$$ក. \lim_{x \rightarrow 0} \frac{\ln(1 + 4044x)}{2x}$$

$$ខ. \lim_{x \rightarrow 0} \frac{\ln(1 - 2023x)}{x}$$

$$គ. \lim_{x \rightarrow 0} \frac{\ln(\cos 2x)}{x^2}$$

$$ឃ. \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{(\pi - 2x)^2}$$

$$ង. \lim_{x \rightarrow 0} \frac{\ln(1 + 12x)}{e^{3x} - 1}$$

$$ច. \lim_{x \rightarrow 0} \frac{e^x - e^{2x}}{\ln(1 + 3x)}$$

$$ឆ. \lim_{x \rightarrow 0} \frac{\tan 2x + \ln(x+1)}{e^{-x} - 1}$$

$$ជ. \lim_{x \rightarrow +\infty} \frac{1}{x} [\ln(x+1) - \ln(x-1)]$$

$$ណ. \lim_{x \rightarrow 0^+} \frac{x^2 + 1}{\ln x + 2}$$

$$ញ. \lim_{x \rightarrow +\infty} \ln \left(\frac{2x+1}{x+1} \right)$$

$$ដ. \lim_{x \rightarrow +\infty} \frac{\ln(x^2 + 4)}{\ln(2x+1)}$$

$$ប. \lim_{x \rightarrow +\infty} \frac{e^x \ln x + 1}{x^2}$$

$$ខ. \lim_{x \rightarrow +\infty} \frac{e^x + \ln x}{x^3}$$

$$ឆ. \lim_{x \rightarrow 0^+} \frac{5 - 3 \ln x}{\ln x + 2}$$

$$ណ. \lim_{x \rightarrow +\infty} \frac{\ln^2 x + 5}{2x^2 - 3}$$

$$គ. \lim_{x \rightarrow +\infty} \frac{\ln(2 + e^{3x})}{\ln(1 + e^{2x})}$$

$$ច. \lim_{x \rightarrow 0^+} \frac{x^2 - 1}{x^3 + \ln x}$$

$$ទ. \lim_{x \rightarrow 0^+} \frac{e^x \ln x + 1}{x^2}$$

$$គ. \lim_{x \rightarrow 0} \frac{\ln(1 + \tan 2x)}{x}$$

$$ស. \lim_{x \rightarrow 0} \frac{\ln(1 + \sin 3x)}{x}$$

$$ប. \lim_{x \rightarrow -\infty} \frac{2 \ln x - 1}{\ln x + 5}$$

$$ឆ. \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\ln(1+x)}$$

$$ព. \lim_{x \rightarrow 0} \frac{2018e^{2018x} + e^{2019x} - 2019}{\ln(1+2018x)}$$

លំហាត់ស្រាវជ្រាវ

1. គណនាលីមីតនៃកន្សោមខាងក្រោម៖

ក. $\lim_{x \rightarrow 2} (3x^2 + 4x + 4)$

ខ. $\lim_{x \rightarrow 4} (5x^2 + 2x - 10)$

គ. $\lim_{x \rightarrow -1} (9x^3 + 3x + 2)$

ឃ. $\lim_{x \rightarrow 3} (3x^2 + 4x + 4)(x^2 + 4)$

ង. $\lim_{x \rightarrow 1} (8x + 1)^2 (3x - 4)^2$

ច. $\lim_{x \rightarrow -1} \sqrt{x^2 - 10x + 5}$

ឆ. $\lim_{x \rightarrow 1} \sqrt[3]{x^2 + 4x + 3}$

ជ. $\lim_{x \rightarrow -1} \frac{(x-1)^2}{x+2}$

ឈ. $\lim_{x \rightarrow -1} \frac{5x^2 - 10x + 8}{x^2 + 9x - 1}$

ញ. $\lim_{x \rightarrow 4} \frac{\sqrt{x^2 + 4x + 4}}{x^2 + x + 2}$

2.

ជូនពរដល់គ្រូការសិក្សា